Completing the square

1 Explain how to find the number to add to $x^2 - 5x$ to make it a perfect square trinomial.

2 Consider the expression $x^2 + 6x$.

a Does adding 36 to the expression make it a perfect square?

b What should be added to $x^2 + 6x$ to make it a perfect square?

3 Consider the equation $x^2 - 3x = 6$. What should be added to the equation in order to complete the square?

4 Complete the following expressions so they form a perfect square:

a $x^2 + 10x + \square$ **b** $x^2 - 2x + \square$ **c** $x^2 - \square x + 16$ **d** $x^2 - x + \square$

e $x^2 + mx + \square$ **f** $x^2 - \frac{4}{5}x + \square$

5 Complete the following perfect squares:

a $x^2 + 4x + \square = (x + \square)^2$ **b** $x^2 - 5x + \square = (x - \square)^2$

c $x^2 - \frac{7}{4}x + \square = (x - \square)^2$

6 Rewrite the following in the form $(x + b)^2 + c$ using the method of completing the square:

a $x^2 + 12x$ **b** $x^2 - 12x$ **c** $x^2 + 2x + 2$ **d** $x^2 - 14x + 56$

e $x^2 - 6x + 5$ **f** $x^2 + 3x + 6$ **g** $x^2 - 7x + 14$

7 Factor the quadratics using completing the square:

a $y = x^2 + 4x + 3$ **b** $y = x^2 - 6x + 8$

c $y = x^2 + 17x + 70$ **d** $y = x^2 - 11x + 28$

e $y = x^2 + 56x + 159$ **f** $y = x^2 - 24x + 63$

8 Factor the quadratic $y = (x + 3)(x + 9) - 7$ using the method of completing the square to get it into the form $y = (x + a)(x + b)$.

9 Factor the quadratics using the method of completing the square to get them into the form:

$$y = \left(x - a + \sqrt{b}\right)\left(x - a - \sqrt{b}\right)$$

a $y = x^2 - 6x + 7$ **b** $y = x^2 + 8x + 14$

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10 Christa attempted to complete the square on the quadratic $y = x^2 - 38x + 105$. Her work is shown below:

1. $y = x^2 - 38x + (-19)^2 - (-19)^2 + 105$
2. $y = (x - 38)^2 - (-19)^2 + 105$
3. $y = (x - 38)^2 - 256$
4. $y = (x - 38)^2 - 16^2$

a What was Christa's mistake?

b Correctly complete the square on the quadratic.

Non-monic completing the square

11 Using the method of completing the square, rewrite $4x^2 - 36x + 79$ in the form $c\left((x + a)^2 + b\right)$.

12 Rewrite the following in the form $a(x + b)^2 + c$ using the method of completing the square:

a $3x^2 + 9x + 8$

b $4x^2 - 11x + 7$

13 Factor the quadratics using completing the square:

a $y = 2x^2 + 25x + 23$

b $y = 4x^2 + 44x + 72$

Quadratic equations

14 Consider the equation $x^2 + 24x = 10$. To solve the equation by completing the square, state whether each of the following lines of work could be used:

a $x^2 + 24x + 144 = 10$

b $(x + 12)^2 - 144 = 10$

c $(x + 12)^2 = 10$

d $(x + 12)^2 - 144 = -134$

e $x^2 + 24x + 144 = 10 + 144$

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15 Solve the following quadratic equations:

a $(x+6)^2 = 4$

b $(4x-4)^2 = 4$

c $2(x-3)^2 = 8$

d $(x+2)^2 = 20$

e $(x+5)^2 - 2 = 15$

f $(x+9)^2 = \frac{15}{2}$

16 Solve the following quadratic equations by completing the square:

a $x^2 + 18x + 32 = 0$

b $x^2 - 6x + 8 = 0$

c $x^2 - 9x + 8 = 0$

d $2x^2 - 12x - 32 = 0$

e $4x^2 + 11x + 7 = 0$

f $x^2 - 2x - 32 = 0$

17 Solve the following quadratic equations by completing the square. Leave your answers in radical form:

a $x^2 + 24x + 5 = 0$

b $x^2 + 11x + 5 = 0$

c $6x^2 + 48x + 24 = 0$

d $x^2 - 7x + 8 = 0$

e $x^2 + \frac{x}{3} - 3 = 0$

f $5x^2 + 55x + 3 = 0$

Applications

18 A circle has equation $(x+2)^2 + (y-4)^2 = 41$. Find the x-intercepts of the circle.

19 Find the value(s) of k that will make $x^2 + kx + 16$ a perfect square trinomial.

20 The revenue y (in millions of dollars) of a company x years after it first started is modeled by

$$y = 12.5x^2 - 64x + 135$$

By completing the square, use this equation to predict the number of years x after which the revenue of the company will reach $\$1167$ million.

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